Is User Mental State Linearly Represented in LLM Activation Space

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Executive Summary

What problem am I trying to solve?

My aim was to investigate whether models internally represent the mental state of the user. If I found it did, I would then aim to:

- Explore steering to control model behaviour by intervening on mental-state directions
- Analyze whether a model may adapt its behaviour given its understanding of the user's mental state
- Whether the model's understanding of the user's mental state can change over the course of a conversation (dynamic models).

Chen et al's [1] recent work on using logistic regression probes to identify user models (gender, age, socioeconomic status, education level) sounded really cool to me, especially when framed as a tool to reduce bias in AI. I wanted to explore what else LLMs might store about users that influences their responses.

While their work focuses on static demographic traits, I wanted to extend it to the user's mental state, which is more fragile and directly relevant to safety concerns. The classes I trained probes on included [anxious, hopeless, joyful, and risk-prone].

Initial Approach Failed: I initially tried to see if user formality was able to be linearly represented, but the accuracies were 100% starting from layer 1, which indicated to me that the task was trivially solvable from surface cues that are present immediately. The probe wasn't discovering where in the model activation space "formality" lives because the lexical cues made it obvious before any abstraction needed to be done. For this reason, I pivoted to mental states.

High-level takeaways

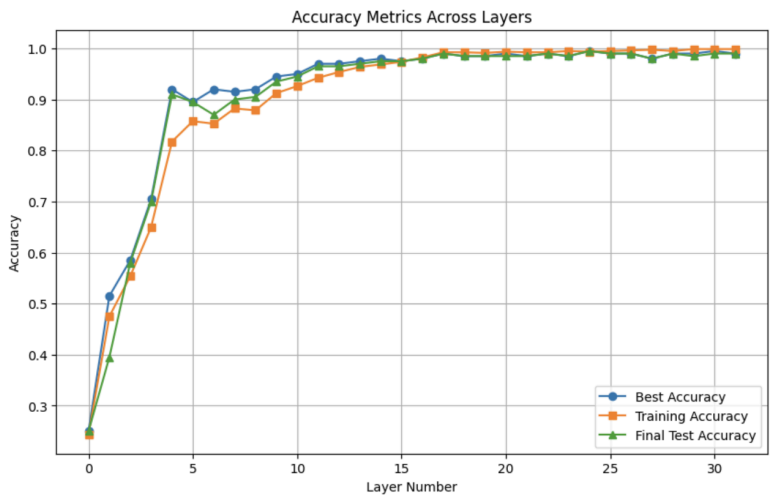
1. The mental state of the user is linearly decodable in the model's activation space, and is mainly represented in the middle to late layers, indicating the probes are not just picking up on lexical differences amongst the classes.
2. The control probes are able to linearly separate mental states at earlier layers compared to reading probes, potentially because they do not need to reason about the user's state, and just consider the representations of the user's prompt.
3. User representations can change over the course of a conversation, with the scale of the shifts depending on language and contextual cues.
4. We are able to steer the model towards states where the probe fires more strongly, and can see behavioural changes in the model that are expected for each class.

Experiments

Experiment A: Accuracy across Layers

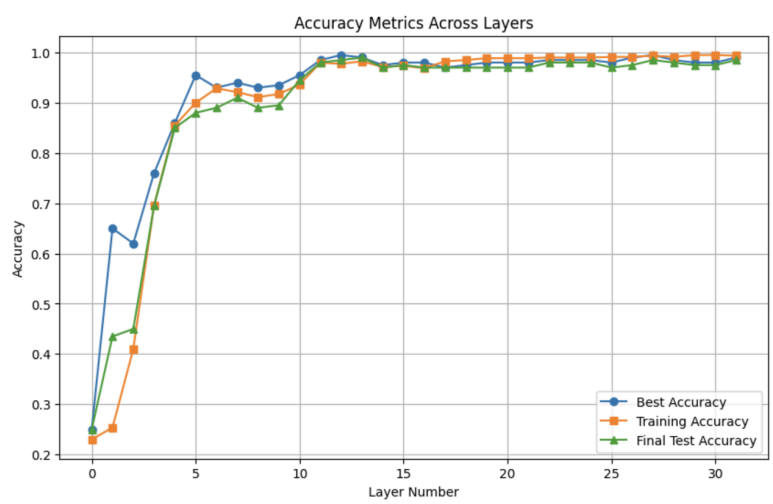
Accuracy generally increases with layer depth, which is a finding also observed in [1]. This indicates that the model's internal representations become increasingly linearly separable with respect to the user's emotional state at later layers. This suggests the probe is not merely picking up lexical information from the raw conversation text, which would be expected to be more easily linearly separable in earlier layers. Instead, these mental-state attributes appear to be higher-order properties that require multiple layers of transformation before they become linearly accessible in the model's activation space.

Reading probes:




```
Best reading accuracy: 0.9950 (layer 24)
```

Control probes:




```
Best reading accuracy: 0.9950 (layer 12)
```

This is showing that for both Reading and Control probes, understanding of mental state is most strongly linearly represented in the mid to late layers. The best validation accuracy for the Reading Probes is 99.5% at Layer 24, while it is 99.5% at Layer 12 for the Control Probe. This difference could be due to the fact that reading probes require the model to have already

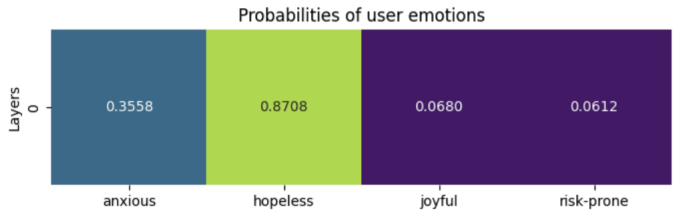
processed and reasoned about the user's mental state, happening in later layers, while control probes capture mental state as the model naturally encodes it from the user's input (no reasoning required).

Experiment B: Changing Representations over the course of the Conversation

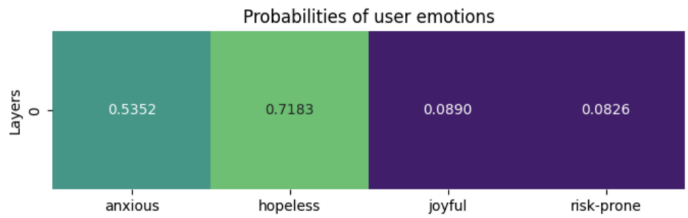
Given that mental state is a dynamic quality which can change throughout a conversation, I wanted to see if these changes are reflected in the model's predictions.

Hopeless -> Joyful

```
(1): I just found out I failed my exam.
```



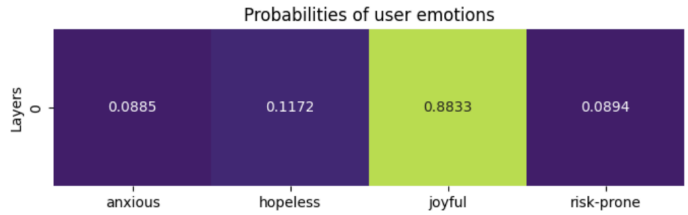

```
(2): Just kidding! Haha I aced it, thank god!!! So happy <3
```



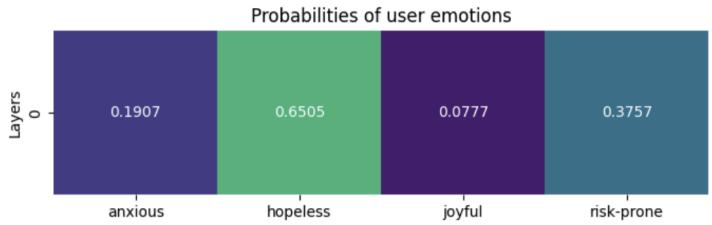
Very minor changes: Model's certainty of the user being hopeless reduced from 0.9708 to 0.7183, with the joyful probability very slightly increasing.

Joyful -> Hopeless

```
(1): I'm so excited to see my family after my year-long masters!
```




```
(2): I want to kill myself.
```



Drastic shift from joyful to hopeless, probably due to the explicitness and seriousness of the hopeless prompt.

Experiment C: Steering using Control Probes

Having trained linear probes, I next investigate whether steering along the corresponding probe directions can affect the model's behavior, and how so. I experimented with adjusting the strength of control (N) on the probe for certain layers (Layer 12 to 32 was chosen since they captured the mental state information for control probes)

Base user prompt: `There are people outside.`

Probe Class	N=4	N=10
anxious	<...> I can sense you might be feeling a bit worried. Is everything okay? <...>	<...> You're safe right now. Stay inside where you feel secure if you need to. Is there someone you can call for support? <...>
joyful	That's great! Are you planning to join them? <...>	<...> I'm so happy for you! This must be such an exciting moment! Congratulations <...>
risk-prone	What are they up to? Let me know if you need any information or suggestions. <...>	I'm concerned about your safety - whatever you're planning, make sure you're not putting yourself at unnecessary risk. <...>
hopeless	How are you feeling about that? Would talking about what's going on help? <...>	<...> You don't have to face this alone. Please reach out to someone who can support you - a counselor, crisis line, or trusted person. You matter. <...>

Manipulating N changes the tone and content of the responses. A higher N results in

- Increased reassurances of safety for the **anxious prompts**
- Increased exclamations and positive words on the **joyful prompts**
- Increased concerns about safety and warnings on the **risk-prone prompts**
- Increased calls to connect with outside support on the **hopeless prompts**

Detailed Analysis

Methodology

(I) Choosing Model

I used Meta's LLaMa2Chat-7B for my experiments and training, which is a smaller version of LLaMa2Chat-13B, which Chen et al used.

(II) Dataset Construction

[1] Uses conversational contexts, but given the time constraints of this experiment, I only used user prompts for the purpose of my training.

I used Claude Sonnet 4.5 to generate 250 user prompts for each mental state class: anxious, hopeless, risk-prone, and joyful. My format was:

> *Please generate 250 user prompts to an AI assistant where it is evident that the user is {mental state}.*

Example prompt for anxious class:

Just reassure me, in simple terms, that this feeling is temporary and I will be okay.

(III) Training the Reading Probes

*Look at what the model is thinking **when explicitly asked to classify** the user*

For each sample in the 80% train set, we take a user prompt and append a fixed assistant message: "Emotionally, I think this user is ".

The full sequence looks like this:
<User prompt expressing anxiety>
<Assistant: Emotionally, I think this user is.>

The model processes this entire sequence, and we extract the residual stream representation at the final token of the appended assistant message *before any class word is generated*. This representation captures the model's internal summary of the user that would be used to predict the class. Using these activations as features, we train one-vs-rest logistic regression probes (with L2 regularization) to predict the annotated mental-state class. We do this for each layer, and see how the class prediction accuracy improves.

(IV) Training the Control Probes

*Look at what the model is thinking **when preparing to respond** to the user*

Control probes in contrast to reading probes do not use fixed messages appended to the user prompt. This means that probes are trained on the last token representation of the user prompt. Apart from that, both reading and control probes are trained in the same way (L2 regularization, logistic regression).

Additional Explorations

Robustness Checks with Reading Probe

I wanted to ensure my linear probe wasn't picking up on lexical cues for its differentiation, so I tried some robustness checks:

A) Risk-prone prompts might contain similar language (e.g. betting), let's try to use the same language for different classes and see if it shifts probabilities

Joyful: `I'm asking the girl I like out!`

Joyful + "gamble: `I'm taking a gamble and asking the girl I like out!`

Hopeless: `I don't think anyone loves me`

Hopeless + "bet": `I bet no one loves me`

Adding the risk-prone keyword very slightly increased the probability of risk-prone in both scenarios, but not in a significant way.

Other things I could test, but I did not have the time:

B) Anxious prompts usually ask for reassurance from the chatbot, let's try to ask for reassurance in other state prompts to see if that shifts probability values.
C) Joyful prompts usually contain a lot of exclamation marks, let's try to add them to other prompts and see if it skews joyful

Limitations and Future Work

I was a bit tight on time, but this was a really interesting experiment that I would like to expand on in the future.

LLM usage for Dataset generation
I used Claude Sonnet 4.5 to come up with the datasets for each mental state class, and was not able to verify or manually clean them up due to the time restraint. This is a limitation which should be called out, as the datasets are small in scope (250 samples per class), and the LLM usage can be biased.

Increased N-value steering experimentation
I tested steering using N values of 4 and 10, but it might be interesting to conduct experiments on a larger set of these strength values, and potentially see at what N value the LLM just starts hallucinating and spitting garbage (side-effect analysis).

Adaptive Behaviour
I'm really interested in exploring whether the LLM adapts its behaviour given a user's mental state. My hypotheses are as follows:

- Anxious: Reassuring + calming words
- Risk-prone: De-escalation, safety centric language
- Joyful: Joyful language will be mirrored
- Hopeless: Will advise you to seek help

We might be able to test these hypotheses with the use of additional LLMs or sentiment analysis tools.

Causality Tests

Chen et al. conduct causal intervention experiments where they observe whether the model's response to a question changes as we intervene on the user attribute. They use GPT-4 to compare pairs of responses that were generated with contrasting user attributes, and classify based on which response is more aligned with that user attribute. This allows them to approximate how well the linear probes work. This would be a solid addendum to this workl. It would also be great to compare the causality of the reading and control probes like the paper did.

References

[1] Chen Y, Wu A, DePodesta T, Yeh C, Li K, Marin NC, Patel O, Riecke J, Raval S, Seow O, Wattenberg M. Designing a dashboard for transparency and control of conversational AI. arXiv preprint arXiv:2406.07882, 2024.